

Branch: gh-pages genstats / inst / doc / 02_11_many-regressions.R

jtleek 14 days ago big update

1 contributor

95 lines (78 sloc) 3.11 KB

Raw

Blame

History



```
1 ## ----global_palette, results = 'asis'-----
2 rm(list=ls())
3 tropical= c('darkorange', 'dodgerblue', 'hotpink', 'limegreen', 'yellow')
4 palette(tropical)
5
6 ## ----global_options,warning=FALSE,message=FALSE-----
7 ## see ch. 10 Hooks of Xie's knitr book
8 library(knitr)
9 knitr_hooks$set(setPch = function(before, options, envir) {
10   if(before) par(pch = 19)
11 })
12 opts_chunk$set(setPch = TRUE)
13
14 ## ----global_plot,warning=FALSE, message=FALSE-----
15 knitr::opts_chunk$set(fig.width=5, fig.height=5, size="footnotesize",
16   warning=FALSE, message=FALSE)
17 knitr::knitr_hooks$set(small.mar = function(before, options, envir) {
18   if (before) graphics::par(mar = c(5,5,1.5,1))
19 })
20
21 ## ----load_hidden, echo=FALSE, results="hide", warning=FALSE-----
22 suppressPackageStartupMessages({
23   library(devtools)
24   library(Biobase)
25   library(limma)
26   library(edge)
27 })
28
29 ## ----load-----
30 library(devtools)
31 library(Biobase)
32 library(limma)
33 library(edge)
34
35 ## ----install_packages, eval=FALSE-----
36 # install.packages(c("devtools"))
37 # source("http://www.bioconductor.org/biocLite.R")
38 # biocLite(c("Biobase", "limma", "jdstorey/edge"))
39
40 ## -----
41 con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bottomly_eset.RData")
42 load(file=con)
43 close(con)
44 bot = bottomly.eset
45 pdata=pData(bot)
46 edata=as.matrix(exprs(bot))
47 fdata = fData(bot)
48 ls()
49
50 ## -----
51 edata = log2(as.matrix(edata) + 1)
52 edata = edata[rowMeans(edata) > 10, ]
53
54 ## -----
55 mod = model.matrix(~ pdata$strain)
56 fit = lm.fit(mod,t(edata))
57 names(fit)
58
59 ## -----
60 fit$coefficients[,1]
61 tidy(lm(as.numeric(edata[1, ]) ~ pdata$strain))
62
63 ## -----
```

```
64 par(mfrow=c(1,2))
65 hist(fit$coefficients[1,],breaks=100,col=2,xlab="Intercept")
66 hist(fit$coefficients[2,],breaks=100,col=2,xlab="Strain")
67 abline(v=0,lwd=3,col=1)
68
69 ## -----
70 par(mfrow=c(1,2))
71 plot(fit$residuals[,1],col=2)
72 plot(fit$residuals[,2],col=2)
73
74 ## -----
75 mod_adj = model.matrix(~ pdata$strain + as.factor(pdata$lane.number))
76 fit_adj = lm.fit(mod_adj,t(edata))
77 fit_adj$coefficients[,1]
78
79 ## -----
80 fit_limma = lmFit(edata,mod_adj)
81 names(fit_limma)
82 fit_limma$coefficients[1,]
83 fit_adj$coefficients[,1]
84
85 ## -----
86 edge_study = build_study(data=edata,grp=pdata$strain,adj.var=as.factor(pdata$lane.number))
87 fit_edge = fit_models(edge_study)
88 summary(fit_edge)
89 fit_edge@beta.coef[1,]
90 fit_limma$coefficients[1,]
91
92 ## ----session_info-----
93 devtools::session_info()
94
```

